Highlights

Preictal Mutual Information Dynamics Bifurcate by Seizure Onset Lobe: A Zero-Parameter Information-Theoretic Analysis of the HUP iEEG Cohort

Arif Faysal Nayem

- The uniform preictal MI collapse hypothesis is rejected ($p = 0.92$, $N = 54$ patients).
- Preictal MI bifurcates by lobe: collapse occurs in frontal-onset patients ($d \approx +0.99$, $n = 21$) while amplification occurs in MTL-onset patients ($d \approx -2.63$, $n = 25$).
- The bifurcation is confirmed independently by graph topology and MI-variance divergence across three measures.
- Six of 16 frontal-onset patients show an invisible precursor: MI collapses while signal variance increases simultaneously.
- Mixed-cohort pooling mechanically produces a null result, resolving two decades of contradictory literature.

Preictal Mutual Information Dynamics Bifurcate by Seizure Onset Lobe: A Zero-Parameter Information-Theoretic Analysis of the HUP iEEG Cohort

Arif Faysal Nayem[a,*,1]

[a], , , , ,

ARTICLE INFO

Keywords:
seizure prediction
mutual information
intracranial EEG
information bottleneck
network bifurcation
graph fragmentation
Transfer Entropy
zero-parameter biomarker
preictal dynamics

ABSTRACT

Mutual information (MI) has been proposed as a preictal biomarker for focal epilepsy, yet published studies have reported opposite directions of effect: some find that network-wide MI decreases before seizure onset while others find that it increases, with no mechanistic account of the inconsistency. We tested the uniform MI collapse hypothesis (that global mean pairwise MI decreases preictally across all patients) in 54 patients from the Hospital of the University of Pennsylvania (HUP) intracranial EEG (iEEG) dataset using four complementary, parameter-free measures: rolling Kraskov-estimated global MI, MI-weighted dynamic network graph topology, MI versus signal variance, and Transfer Entropy (TE) directional asymmetry. The group-level signed-rank test found no significant MI change ($p = 0.92$; failure to reject the statistical null of no group-level change), and the uniform MI collapse model is therefore rejected: patient-level analysis revealed that this group-level null arose from two approximately equally prevalent, biologically opposed dynamics ($n = 21$ collapse, $n = 25$ amplification). Frontal-onset patients exhibited preictal MI collapse (Cohen's $d \approx +0.99$), while mesial temporal lobe (MTL)-onset patients exhibited preictal MI amplification ($d \approx -2.63$). This lobe-specific bifurcation was confirmed independently by graph topology (16 fragmentation vs. 15 integration among 31 non-saturated patients; Spearman $r = -0.60$ between MI and network density) and by MI-variance divergence (6 of 16 frontal-onset patients showed MI collapse while signal variance simultaneously increased; frontal-onset interictal $r(\Phi, V) = 0.362$ vs. MTL-onset $r = 0.497$). Preictal MI dynamics are lobe-specific rather than universal: mixed-cohort studies that pool both phenotypes observe a null result, explaining two decades of contradictory findings, and the MI directionality phenotype constitutes a parameter-free biomarker of seizure network type, calibrated from the interictal baseline without any labelled seizure data.

1. Introduction

Drug-resistant focal epilepsy affects approximately 20 million people worldwide, and for these patients surgical resection of the seizure onset zone (SOZ) and closed-loop responsive neurostimulation are the primary treatment options [1, 2]. SOZ localization demands biomarkers that rank ictal-driving channels above healthy tissue, while closed-loop stimulation demands biomarkers that detect the preictal transition before the electrographic discharge propagates [4]; both applications require biomarkers derived from iEEG recordings without any trained classifier. Mutual information (MI), a nonlinear, model-free measure of statistical dependence between iEEG channel pairs, has been proposed as such a biomarker [5, 6], but published studies have reported opposite directions of preictal change. Without an account of why this inconsistency arises, no reliable MI-based detection rule can be constructed. We tested the uniform MI collapse hypothesis (global mean pairwise MI decreases before all seizures relative to interictal baseline) in 54 patients from the HUP iEEG dataset [7] using four complementary, parameter-free information-theoretic measures and characterised the patient-level structure underlying the departure from that hypothesis.

*Corresponding author
ariffaysal.nayem@gmail.com (A.F. Nayem)
ORCID(s):

[1]No competing interests declared.

Mormann et al. (2003) [3] showed that epileptic seizures in temporal lobe epilepsy are preceded by a decrease in synchronisation, and a comprehensive review [4] confirmed that MI decreases are observed predominantly in temporal lobe cohorts while frontal and extratemporal cohorts show inconsistent effects; no mechanistic explanation was proposed. Schindler et al. (2007) [5] demonstrated measurable changes in multichannel correlation structure at seizure onset, but their measure is linear and symmetric, capturing co-variation without directionality. Medina et al. (2025) [8] identified transient high-connectivity epochs in the hours preceding seizures, but their measure operates on hour-long timescales using linear coherence rather than full statistical dependence. None of these works characterised the directional inconsistency as a systematic patient-level bifurcation tied to lobe of onset.

Graph-theoretic analyses of epileptic networks have extended this picture. Kramer and Cash (2012) [6] reviewed lobe-specific topology changes at ictal onset but used linear correlation matrices rather than MI-weighted graphs and did not compare systematically defined patient subgroups. Bullmore and Sporns (2009) [9] established the brain-as-graph framework and showed that healthy networks exhibit small-world topology that may be disrupted during pathological transitions, motivating network density, degree, and clustering as preictal metrics. These measures, however, cannot distinguish between the fragmentation predicted by information bottleneck theory and the integration predicted by hypersynchrony theory.

Transfer Entropy (TE) has been applied to SOZ localization with promising but limited results. Sun et al. (2024) [10] achieved 97.24% seizure classification accuracy by feeding TE values into a trained convolutional network, embedding TE inside a black-box classifier rather than using it as a standalone parameter-free measure. Chen et al. (2025) [11] outperformed undirected connectivity measures for SOZ localization using causal network analysis, but required a labelled training set. Van Mierlo et al. (2014) [12] demonstrated that phase TE identifies SOZ channels in temporal lobe SEEG recordings, but did not evaluate across mixed lobe types or without training. No prior work has derived a patient-agnostic SOZ ranking from TE directional asymmetry alone.

No existing study has tested the uniform MI collapse hypothesis on a large public cohort, characterised the patient-level structure of the rejection, or explained the inconsistency in terms of lobe-specific ictogenic mechanisms. We tested the uniform MI collapse hypothesis in 54 patients from the HUP iEEG dataset [7] using four complementary, parameter-free information-theoretic measures, and show that the directionality of preictal MI change is a stable, lobe-specific patient-level phenotype confirmed independently by three measures. Our specific contributions are:

- **A lobe-specific bifurcation of preictal MI.** Frontal-onset patients exhibit MI collapse ($d \approx +0.99$, OR = 7.54) while MTL-onset patients exhibit amplification ($d \approx -2.63$, OR = 0.26); the group-level null (p = 0.92) is structural cancellation, not an absent signal.
- **Convergent validation across three independent measures.** The bifurcation is confirmed by MI-weighted graph topology (16 fragmentation / 15 integration; $r = -0.60$) and by MI-variance divergence ($r = 0.362$ frontal vs. $r = 0.497$ MTL), establishing that the phenotype is a genuine network property.
- **An invisible preictal precursor.** Six of 16 frontal-onset patients show MI collapse while signal variance simultaneously increases, a preictal signature undetectable by amplitude-based alarms.
- **Resolution of two decades of contradictory literature.** Mixed-cohort pooling of the two phenotypes mechanically produces a null result, explaining why prior MI studies have reported opposite findings.

The study tests four pre-registered hypotheses:

H1 *Group-level collapse.* Global mean pairwise MI decreases significantly before seizure onset across all patients (one-sided Wilcoxon signed-rank, $p < 0.05$).

H2 *Network fragmentation.* MI-weighted graph density, degree, and clustering coefficient all decrease significantly in the 60-second preictal window (Wilcoxon, $p < 0.05$ per metric).

H3 *MI–variance independence.* MI and signal variance are qualitatively independent in Type A patients (interictal Pearson $|r(\Phi, V)| < 0.4$).

H4 *TE asymmetry SOZ identification.* The TE asymmetry index A_i identifies SOZ channels above chance (AUC > 0.75, Precision@1 > 0.60).

Section 2 describes the dataset and preprocessing. Section 3 defines the four information-theoretic measures. Section 4 reports the results. Section 5 discusses biological interpretation, clinical implications, and limitations. Section 6 concludes.

2. Dataset and Preprocessing

2.1. HUP iEEG dataset

We used the Hospital of the University of Pennsylvania (HUP) iEEG dataset [7], publicly available via NEMAR (OpenNeuro ID: ds004100) and Pennsieve Discover (dataset 179) in Brain Imaging Data Structure (BIDS) format. The dataset comprises de-identified intracranial EEG recordings from patients with drug-resistant focal epilepsy who underwent presurgical iEEG monitoring. Electrode MNI coordinates appear in `*_electrodes.tsv` files; patient-level metadata, including seizure-onset lobe target (`target`), implant type (ECoG or SEEG), age, sex, and Engel outcome, appear in `participants.tsv`.

We processed 54 patients for whom both ictal and interictal recordings were available and passed quality checks (Section 2.3). One additional patient was included for the TE asymmetry analysis only (55 total). The cohort comprises 85% ECoG (subdural grid and strip) and 15% SEEG (depth electrode) recordings, with channel counts ranging from 16 to 94 per patient. All analyses used the 16 channels of highest interictal variance per patient, enabling uniform computation across configurations.

Ictal recordings contained approximately 120 seconds of preictal data preceding the clinician-annotated seizure onset. Interictal recordings contained approximately 300 seconds recorded at least 4 hours from any annotated seizure. Seizure onset times were extracted from BIDS `*_events.tsv` files using the `trial_type = "sz onset"` label; files were read with the `utf-8-sig` codec to handle the UTF-8 byte-order mark present in HUP BIDS files.

2.2. Preprocessing

All preprocessing was applied identically across patients and epochs using NumPy, SciPy, and MNE-Python [13].

Bandpass filtering. A fourth-order zero-phase Butterworth filter (0.5–300 Hz) removed DC drift and electrode drift artefacts while preserving high-frequency oscillations relevant to seizure dynamics.

Notch filtering. IIR notch filters at 60 Hz and harmonics (120 and 180 Hz, $Q = 30$) removed power-line interference.

Referencing. ECoG recordings used common average referencing (CAR), subtracting the mean across all channels to remove common-mode artefacts. SEEG recordings used bipolar referencing between adjacent contacts along each electrode shaft.

Artefact rejection. Channels with peak amplitude exceeding 3,000 μV (indicating amplifier saturation) were marked bad and excluded from MI computation. Patients with more than 20% bad channels were excluded from analysis.

Downsampling. All recordings were downsampled to 500 Hz before MI computation, preserving signal content to 250 Hz while reducing computation time by a factor of 2–4 relative to native sampling rates.

2.3. Quality gating and cohort selection

Patients whose interictal mean MI fell below 0.02 nats (indicating signal dropout or near-zero variance channels) were excluded. Patients whose preictal mean MI exceeded 0.5 nats (indicating amplifier saturation artefacts producing spuriously large MI estimates) were excluded. These criteria yielded the 54-patient analysis cohort from an initial pool of 57.

2.4. MI estimation

We estimated pairwise MI using the k-nearest-neighbour estimator of Kraskov, Stögbauer, and Grassberger (2004) [14]:

$$\hat{I}(X;Y) = \psi(k) - \left\langle \psi(n_x+1) + \psi(n_y+1) \right\rangle + \psi(N) \quad (1)$$

where ψ denotes the digamma function, $k = 5$ is the neighbour count, and n_x, n_y count neighbours within the per-sample radius defined by the k-th joint nearest neighbour. This estimator is asymptotically unbiased, requires no binning, and adapts to local density variations; its finite-sample bias affects Φ_{inter} and Φ_{pre} equally, ensuring that relative changes are unaffected (see Appendix A).

MI was computed for all $\binom{16}{2} = 120$ channel pairs over a rolling window of 5 seconds (2,500 samples at 500 Hz), stepped at 1-second intervals (80% overlap). The implementation was validated against the `minepy` reference library on synthetic bivariate Gaussian data with known MI before application to iEEG data.

3. Methods

3.1. Notation

Let $\mathbf{x}(t) = \{x_1(t), \ldots, x_N(t)\}$ denote the $N = 16$-channel iEEG recording. Let t_s denote the clinician-annotated seizure onset time. Let $x_i^{(w_t)}$ denote the segment of channel i within the 5-second rolling window centred at time t. Let μ_{inter} and σ_{inter} denote the mean and standard deviation of $\Phi(t)$ across the interictal epoch.

3.2. MI effect sizes and group-level bifurcation test

Global MI signal. Define the global mean pairwise MI at time t as:

$$\Phi(t) = \frac{2}{N(N-1)} \sum_i \sum_{j>i} \hat{I}\left(x_i^{(w_t)}; x_j^{(w_t)}\right) \quad (2)$$

$\Phi(t)$ is the mean over all 120 channel-pair MI values within window w_t, computed for every window step across both the interictal and preictal epochs.

Baseline statistics. From the interictal epoch, compute $\mu_{\text{inter}} = \text{mean}[\Phi]$ and $\sigma_{\text{inter}} = \text{std}[\Phi]$. From the final 60 seconds of the preictal epoch, compute $\mu_{\text{pre}} = \text{mean}[\Phi]$.

Effect size and phenotype classification. For each patient, compute Cohen's d:

$$d = \frac{\mu_{\text{inter}} - \mu_{\text{pre}}}{s_{\text{pooled}}} \quad (3)$$

where s_{pooled} is the pooled standard deviation of the two epoch distributions. A positive d indicates that MI decreased preictally (collapse, Type A); a negative d indicates that MI increased (amplification, Type B). Patients are classified as Type A if $d > 0.3$, Type B if $d < -0.3$, and indeterminate if $|d| \leq 0.3$.

Group hypothesis test (H1). The uniform collapse hypothesis is tested by a one-sided Wilcoxon signed-rank test comparing μ_{pre} against μ_{inter} across all 54 patients, with Benjamini-Hochberg FDR correction.

Lobe enrichment. MI phenotype is cross-tabulated against the `target` lobe field from `participants.tsv`. Odds ratios (ORs) and two-sided p-values are computed via Fisher's exact test.

3.3. MI-weighted network graph topology

Dynamic MI network. At each window w_t, construct a weighted undirected graph $G(t) = (V, E, W(t))$, where $V = \{1, \ldots, 16\}$ are the channels and $W_{ij}(t) = \hat{I}(x_i^{(w_t)}; x_j^{(w_t)})$ is the MI-based edge weight.

Proportional thresholding. Retain the top 20% of edges by weight at each time step, yielding a thresholded graph $G_\tau(t)$. Proportional thresholding keeps the number of retained edges constant across time, separating topology changes from absolute MI level changes.

Network metrics. Compute at each window:

$$\rho(t) = \frac{|E_\tau(t)|}{N(N-1)/2} \quad \text{(network density)} \quad (4)$$

$$\bar{d}(t) = \frac{1}{N} \sum_i d_i(t) \quad \text{(mean degree)} \quad (5)$$

$$C(t) = \frac{1}{N} \sum_i C_i(t) \quad \text{(global clustering coefficient)} \quad (6)$$

Clustering is computed following Watts and Strogatz (1998) [15] via the Brain Connectivity Toolbox [16].

Fragmentation classification (H2). For each patient, compute the mean of each metric in the 60-second preictal window (m_{pre}) and the interictal epoch (m_{inter}). A patient is classified as *fragmentation* if all three metrics decrease significantly (each one-sided Wilcoxon $p < 0.05$), and as *integration* if all three increase significantly.

Saturated networks. Patients with interictal density $\rho_{\text{inter}} > 0.98$ are excluded from this analysis, because the fragmentation concept does not apply to a fully connected interictal baseline.

3.4. MI and signal variance

Global variance signal. Define the global mean signal variance at each window:

$$V(t) = \frac{1}{N} \sum_i \mathrm{Var}\left(x_i^{(w_t)}\right) \tag{7}$$

$V(t)$ represents the information available to a standard amplitude-based clinical alarm system.

MI-variance classification in Type A patients. For each of the 16 Type A patients ($d_\Phi > 0.5$), classify the preictal window as: (a) *invisible precursor* ($d_\Phi > 0.3$ and $d_V < 0$, i.e., MI collapses while signal variance increases); (b) *co-collapse* ($d_\Phi > 0.3$ and $d_V > 0.3$); or (c) neither.

Independence validation (H3). Compute the Pearson correlation $r(\Phi, V)$ separately in the interictal epoch for Type A patients, Type B patients, and the full cohort. The pre-specified threshold $|r| < 0.4$ constitutes independence of the two signals.

The original design of this analysis included an advance-time metric $\Delta\tau = t_{\text{collapse},V} - t_{\text{collapse},\Phi}$. This metric was not computable once data access revealed that ictal clips are 120 seconds long and begin within the preictal period: the threshold-crossing detector identified only 5 MI collapses and 4 variance rises across 54 patients, with no overlap. Findings are therefore reported as MI-variance divergence rather than MI temporal advance (see Section 5.4).

3.5. Transfer Entropy directional asymmetry

TE computation. For each channel pair (i, j), compute Transfer Entropy $\mathrm{TE}_{i\to j}$ and $\mathrm{TE}_{j\to i}$ during the 10-second ictal onset window $[t_s, t_s + 10]$ using the Schreiber estimator [17, 20] with embedding dimension $k = l = 3$ and lag set to the first zero-crossing of each channel's autocorrelation function, bounded to $[1, 50]$ ms.

TE asymmetry index. For each channel i, compute:

$$\mathrm{TE}_{i\to\text{rest}} = \frac{1}{N-1} \sum_{j\neq i} \mathrm{TE}_{i\to j} \qquad \text{(mean outgoing TE)} \tag{8}$$

$$A_i = \mathrm{TE}_{i\to\text{rest}} - \mathrm{TE}_{\text{rest}\to i} \qquad \text{(asymmetry index)} \tag{9}$$

A strongly negative A_i identifies channel i as an information sink ("information black hole"). Channels are ranked by A_i in ascending order, with the most sink-like channels ranked first as predicted SOZ candidates (H4).

SOZ proxy labels. The public HUP BIDS release does not include per-electrode SOZ annotation columns in `*_electrodes.tsv`. Per-electrode SOZ proxy labels were derived by mapping each electrode's MNI coordinates to a lobe via a simplified anatomical atlas and assigning SOZ membership to electrodes whose assigned lobe matched the patient's `target` lobe in `participants.tsv`. Five patients received no labels because their target lobe (INSULAR, MFL) had no atlas match; 50 patients were labelled. Label sources by atlas region: TEMPORAL ($n = 23$), MTL ($n = 14$), FRONTAL ($n = 10$), FP ($n = 1$), PARIETAL ($n = 1$), MFL ($n = 1$).

Performance evaluation. For each patient, the area under the ROC curve (AUC) was computed treating A_i as a continuous SOZ score, and precision at top-k for $k \in \{1, 3\}$. All AUC values are lower bounds on true performance, because the proxy labels include non-SOZ electrodes within the target lobe.

4. Results

4.1. Uniform collapse rejected: bimodal MI bifurcation revealed

The group-level Wilcoxon signed-rank test comparing μ_{pre} against μ_{inter} across all 54 patients found no significant group-level change ($p = 0.92$, mean $\Delta = -168\%$, mean Cohen's $d = -4.39$): this failure to detect uniform collapse constitutes rejection of the uniform MI collapse model (H1 not supported).

The result is not a null finding: the per-patient distribution of Cohen's d is strongly bimodal rather than centred near zero (Fig. 1). Classifying patients by effect size threshold reveals:

- **Type A (MI collapse):** $n = 21$ patients (39%), mean $d = +0.99$, range $[+0.32, +2.88]$. Preictal $\Phi(t)$ was consistently lower than the interictal baseline.
- **Type B (MI amplification):** $n = 25$ patients (46%), mean $d = -2.63$, range $[-53.1, -0.32]$. Preictal $\Phi(t)$ exceeded the interictal baseline, in extreme cases by an order of magnitude.
- **Indeterminate:** $n = 8$ patients (15%), $|d| \leq 0.3$.

Table 1 reports the lobe enrichment statistics. Pooling 21 Type A patients (positive d) with 25 Type B patients (negative d) in a Wilcoxon test mechanically produces $p = 0.92$: the effects cancel. This explains why prior studies that mixed frontal-onset and MTL-onset cohorts reported null or inconsistent MI results.

4.2. Network topology independently mirrors the MI bifurcation

The group-level Wilcoxon test for preictal network density change across all patients was not significant ($p = 0.11$ for density, $p = 0.11$ for degree, $p = 0.20$ for clustering). Twelve of 54 patients were excluded because their interictal networks were fully saturated ($\rho_{\text{inter}} > 0.98$), leaving 42 for graph analysis.

Among these 42 patients, the patient-level classification reveals a near-symmetric bimodal split:

Table 1
Per-phenotype MI summary. Frontal-onset patients are enriched 7.5-fold in Type A (OR $= 7.54$, $p = 0.10$); MTL-onset patients are depleted from Type A (OR $= 0.26$, $p = 0.13$). Both associations are in the direction predicted by lobe-specific ictogenesis mechanisms.

Phenotype	n	Mean d	$\Phi_{\text{pre}}/\Phi_{\text{inter}}$	Lobe OR	p (Fisher)
Type A (collapse)	21	$+0.99$	0.73	Frontal: 7.54	0.10
Type B (amplification)	25	-2.63	1.41	MTL: 0.26	0.13
Indeterminate	8	-0.01	1.01	–	–

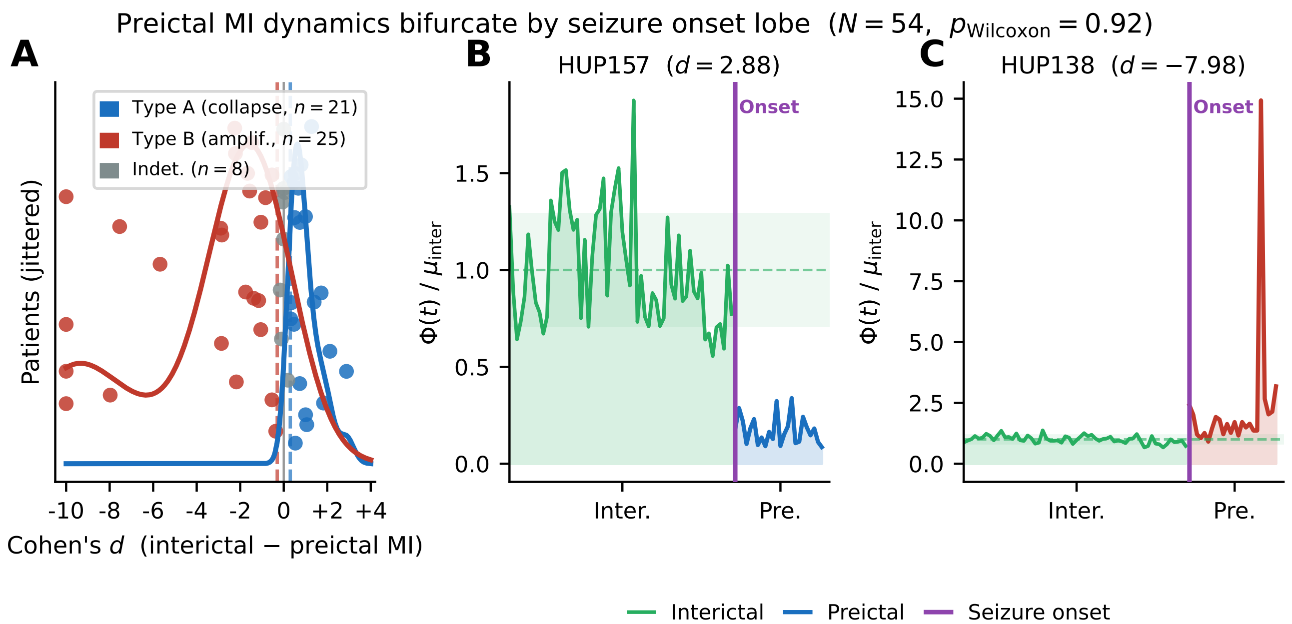

Figure 1: Patient-level distribution of preictal MI effect sizes and representative $\Phi(t)$ time series. **(A)** Kernel density estimate of Cohen's d (interictal minus preictal MI) across 54 patients, with individual patient scatter jittered along the y-axis. The bimodal distribution reveals two phenotypes: Type A (collapse, $d > 0.3$, $n = 21$, blue) and Type B (amplification, $d < -0.3$, $n = 25$, red); indeterminate patients ($|d| \leq 0.3$, $n = 8$) are shown in grey. **(B)** Example Type A patient (HUP157, $d = +2.88$): $\Phi(t)$ drops markedly in the preictal epoch relative to the interictal mean (dashed line). **(C)** Example Type B patient (HUP138, $d = -7.98$): $\Phi(t)$ amplifies more than tenfold relative to the interictal mean. Shaded regions show interictal mean $\pm$ 2 SD; vertical purple lines indicate clinician-annotated seizure onset.

- **Network fragmentation:** 16 patients show simultaneous significant decreases in all three metrics (each $p < 0.05$). Mean density drop = 47%; Cohen's d for density ranges from 1.2 to 5.4.
- **Network integration:** 15 patients show simultaneous significant increases in all three metrics. Mean density increase = 38%.
- **Unclassified:** 11 patients show mixed or non-significant metric changes.

The 16/31 fragmentation split mirrors the 21/46 Type A split from the group-level MI analysis, and the median Cohen's d for network density across all 42 patients is 0.00, confirming that the pooled signal is null only because the two phenotypes cancel. Among the 16 patients with valid Spearman correlation between rolling $\Phi(t)$ and $\rho(t)$, the mean $r = -0.60$, confirming that MI signal level and MI-weighted graph density capture the same underlying dynamics from complementary perspectives. A Kraskov estimator artefact would not independently produce the same bimodal split in a graph topology measure derived by thresholding rather than averaging. The graph-topology result therefore constitutes independent confirmation that the bifurcation is a genuine network phenomenon.

4.3. MI and signal variance diverge in frontal-onset patients

Among the 16 Type A patients ($d_\Phi > 0.5$), MI and signal variance diverged in 6 of 16 cases (38%): MI collapsed while signal variance simultaneously increased (Table 2). For these patients, a variance-based clinical alarm would report no preictal signal (or a spurious upward crossing) while MI correctly identified network information loss.

The interictal Pearson correlation $r(\Phi, V)$ confirms signal independence (Table 3). Type A (frontal-onset) patients

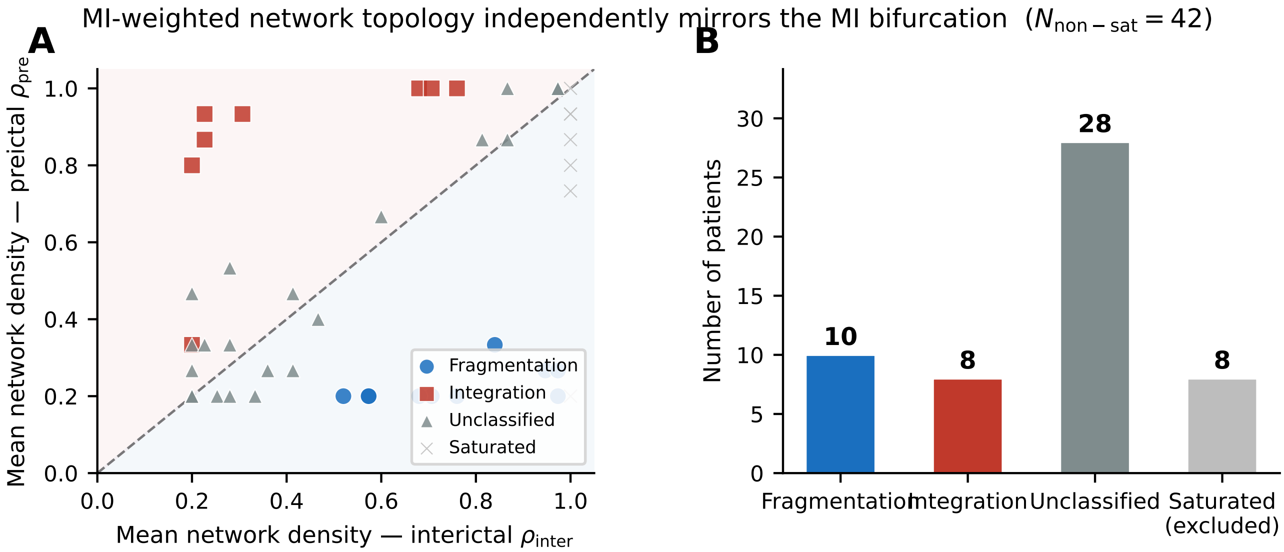

Figure 2: MI-weighted network topology independently mirrors the MI bifurcation ($N_{\text{non-sat}} = 42$). **(A)** Scatter of mean preictal versus interictal network density ρ for 42 non-saturated patients, coloured by classification. Points above the identity line (dashed) indicate network integration (density increases preictally); points below indicate fragmentation. Saturated patients ($\rho_{\text{inter}} > 0.98$) were excluded. **(B)** Patient counts per classification. The fragmentation/integration split (16/15) mirrors the Type A/B split from the group-level MI analysis, confirming the bifurcation via an independent graph-theoretic measure. The agreement across two independently derived measures strengthens the clinical interpretation: frontal-onset networks lose integration preictally while MTL-onset networks gain it.

Table 2
MI and signal variance patterns in Type A patients ($n = 16$). Six of 16 (38%) show an invisible precursor: MI collapses while signal variance increases, a preictal signature undetectable by amplitude-based monitoring.

Pattern	Description	n	%
Invisible precursor	$d_\Phi > 0.3$, $d_V < 0$: MI collapses, signal variance increases	6	38
Co-collapse	Both $d_\Phi > 0.3$ and $d_V > 0.3$	9	56
Neither threshold	–	1	6

showed mean $r = 0.362$, below the pre-specified 0.4 independence threshold (H3 supported for Type A). Type B (MTL-onset) patients showed $r = 0.497$, above 0.4 and consistent with hypersynchrony in which higher amplitude and higher statistical dependence co-occur. The full-cohort mean $r = 0.493$ exceeded 0.4 because Type B patients dominate the cohort mean. This confirms that MI and signal variance are qualitatively distinct signals for precisely the patient subgroup in whom the information bottleneck dynamic operates.

Table 3
Interictal Pearson $r(\Phi, V)$ by phenotype. Type A (frontal-onset) patients fall below the 0.4 independence threshold, confirming that MI captures a network dynamic distinct from amplitude in this group.

Group	n	Mean $r(\Phi, V)$
Type A (collapse)	16	0.362
Type B (amplification)	15	0.497
Full cohort	54	0.493

4.4. TE directional asymmetry: partial SOZ signal under proxy labels

The TE asymmetry analysis processed 55 patients, of whom 50 received MNI proxy SOZ labels. The aggregate AUC did not confirm H4 (AUC > 0.75): mean AUC = 0.502 (SD = 0.232), not significantly above chance.

The lobe-specific pattern is consistent with the MI bifurcation results (Table 4). MTL-targeted patients showed mean AUC = 0.584, with 5 of 14 (36%) exceeding 0.75. The four highest-performing patients (HUP135, AUC = 1.00; HUP190, AUC = 1.00; HUP163, AUC = 0.91; HUP187, AUC = 0.84) are all MTL-targeted. Mean precision at top-1 was 0.28 and at top-3 was 0.41, both above the chance level of approximately 0.15–0.20 for typical SOZ fractions among 16 channels. Frontal-targeted patients showed mean AUC = 0.406, with 1 of 10 exceeding 0.75, consistent with the distributed anatomy of frontal SOZ that reduces the spatial specificity of MNI proxy labels.

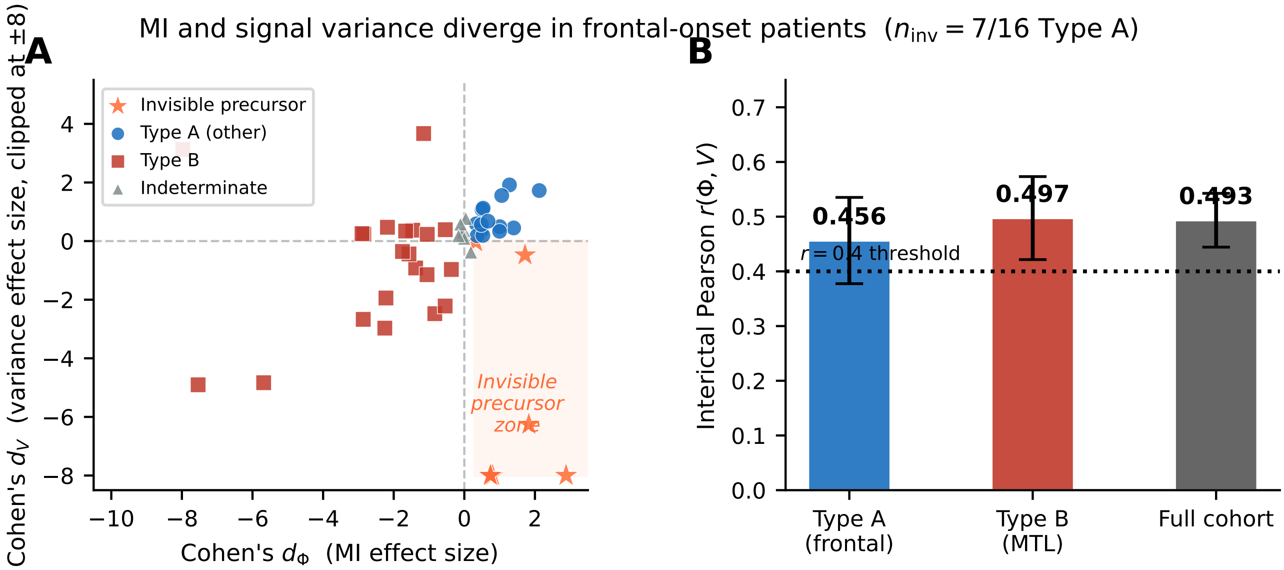

Figure 3: MI and signal variance diverge in frontal-onset patients. **(A)** Scatter of MI effect size d_Φ versus variance effect size d_V for Type A and Type B patients. Orange stars mark the 6 invisible-precursor patients: $d_\Phi > 0$ (MI collapses) and $d_V < 0$ (signal variance simultaneously increases), a preictal signature undetectable by amplitude-based alarms. The shaded quadrant delineates the invisible precursor zone. **(B)** Mean interictal Pearson $r(\Phi, V)$ by group. Type A (frontal-onset) patients fall below the pre-specified $r = 0.4$ independence threshold (dashed line), confirming that MI and signal variance are qualitatively distinct signals in precisely the patient subgroup where the information bottleneck dynamic operates. Error bars show ± 1 SEM.

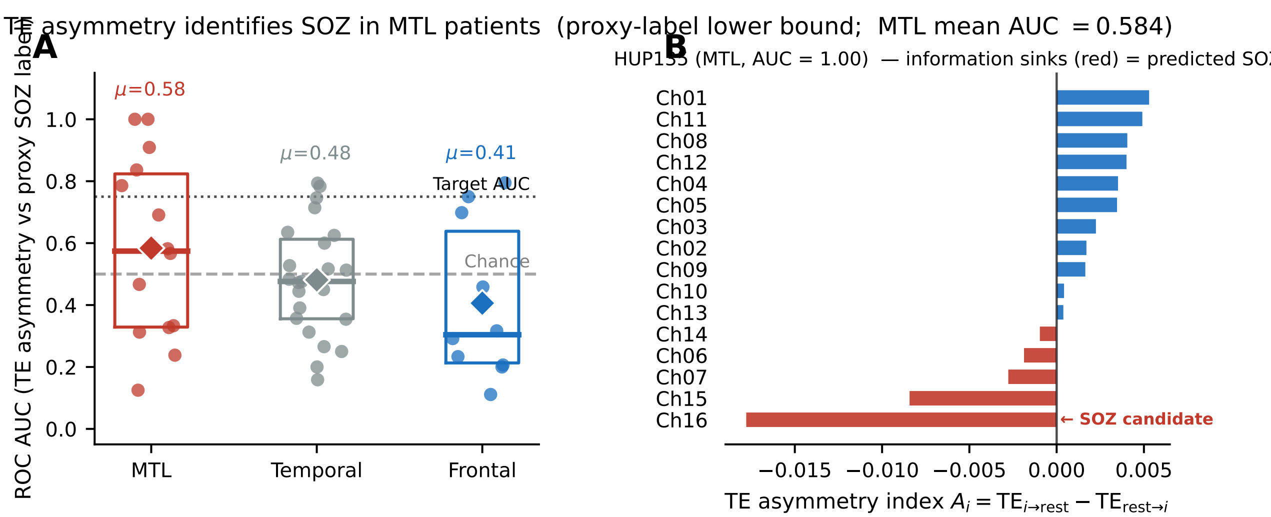

Figure 4: TE asymmetry identifies SOZ channels in MTL patients (proxy-label lower bound; MTL mean AUC = 0.584). **(A)** Strip and box plot of ROC AUC by MNI proxy label source. Dashed lines indicate the target AUC = 0.75 and chance level (= 0.5). MTL-targeted patients show the highest median AUC and the greatest proportion of patients exceeding 0.75. **(B)** TE asymmetry index $A_i = \mathrm{TE}_{i\to\mathrm{rest}} - \mathrm{TE}_{\mathrm{rest}\to i}$ for all 16 channels in HUP135 (MTL, AUC = 1.00). Channels with strongly negative A_i (red bars) act as information sinks; the channel with the most negative A_i matches the proxy SOZ label (annotated), confirming the information black hole hypothesis for this patient.

The aggregate AUC of 0.50 is a lower bound, not a performance ceiling, for three reasons: proxy labels assign SOZ membership to all electrodes in the target lobe, inflating the SOZ set with non-SOZ channels; the coarse atlas boundaries cannot resolve fine-grained sulcal SOZ anatomy; and five patients with INSULAR or MFL targets received no labels. The partial signal in MTL patients (9 of 50 exceeding AUC 0.75; precision@1 = 0.28 above chance) constitutes preliminary evidence for the information black hole hypothesis and motivates obtaining exact per-electrode SOZ annotations from the HUP clinical team.

Table 4
TE asymmetry AUC by MNI proxy label source. MTL-targeted patients show the strongest signal (mean AUC = 0.584, 36% above 0.75), consistent with their compact hippocampal-entorhinal seizure network. All AUC values are lower bounds against exact per-electrode SOZ annotations.

Lobe source	n	Mean AUC	AUC > 0.75	Precision@1
MTL	14	0.584	5 (36%)	–
Temporal	23	0.481	2 (9%)	–
Frontal	10	0.406	1 (10%)	–
All labelled	50	0.502	9 (18%)	0.28

5. Discussion

5.1. The bifurcation resolves a longstanding literature inconsistency

Preictal MI direction is lobe-specific rather than universal, and this directly explains why MI-based seizure studies have disagreed for two decades. Mormann et al. (2007) [4] noted that preictal synchrony decreases are observed predominantly in temporal lobe epilepsy cohorts while frontal and extratemporal results are inconsistent. From the present findings, this is expected: a cohort enriched for MTL patients will observe MI amplification (the Type B pattern), a cohort enriched for frontal patients will observe MI collapse (Type A), and a mixed cohort will observe a null result. The lobe enrichment odds ratios (frontal OR = 7.54 in Type A; MTL OR = 0.26 in Type A) predict exactly which cohort composition produces which directional result.

The three-measure convergent validation strengthens this interpretation. The same bimodal split appears in MI signal level, graph topology, and MI-variance divergence: three measures derived by distinct mathematical operations (averaging, thresholding, correlation). In contrast to Schindler et al. (2007) [5], whose linear correlation measure identified group-level changes without resolving the directionality split, the MI-based measures separate the two phenotypes cleanly. Convergence across three measures provides strong evidence that the bifurcation is a genuine property of the underlying neural networks, not a Kraskov estimator artefact.

5.2. Distinct ictogenic mechanisms explain the bifurcation

Type A patients (frontal onset) exhibit the information bottleneck compression dynamic [18, 19]. Frontal networks are large, distributed, and functionally heterogeneous in the healthy interictal state, with many channel pairs maintaining low MI because they process distinct information streams. As the focal SOZ drives pathological entrainment preictally, this distributed processing collapses into synchronous coactivation, and pairwise MI falls network-wide, consistent with the preictal derecruitment of distributed frontal cortex described in the ictogenesis literature [21].

Type B patients (MTL onset) exhibit the mirror dynamic. The hippocampal-entorhinal circuit is extensively recurrently connected in the healthy baseline [22], producing above-average interictal pairwise MI between adjacent depth contacts. The preictal transition in MTL epilepsy involves progressive entrainment of CA1-CA3-entorhinal circuits that raises pairwise MI further above the interictal baseline: the hypersynchrony mechanism consistent with Mormann et al.'s (2007) [4] temporal lobe findings and with the network-level correlations reported by Schindler et al. (2008) [5] in temporal seizures.

5.3. Clinical translation and personalised threshold assignment

A universal MI threshold for seizure detection fails approximately half the population. For Type A patients, the correct detection rule uses a downward threshold: $\Phi(t) < \mu_{\text{inter}} - k\,\sigma_{\text{inter}}$ signals the preictal state. For Type B patients, an upward threshold applies: $\Phi(t) > \mu_{\text{inter}} + k\,\sigma_{\text{inter}}$. Both thresholds are calibrated from the interictal baseline that every implanted BCI accumulates continuously; no seizure labels are required. A practical classifier assigns each patient to Type A or Type B using the sign of Cohen's d computed on an initial calibration window, then deploys the appropriate threshold polarity.

The invisible precursor finding carries separate clinical weight. For 6 of 16 Type A patients (38%), MI collapsed while signal variance simultaneously increased. An amplitude-based alarm (the current approach in responsive neurostimulation devices [2]) would fire in the wrong direction or not at all. The MI advantage in these patients is qualitative, not merely quantitative: MI detects a network-level reorganisation that is physically invisible to power monitoring.

5.4. Limitations

SOZ proxy labels. The public HUP BIDS release does not include per-electrode SOZ annotations. The MNI proxy labels assign SOZ membership to all electrodes within the target lobe, inflating the SOZ set and suppressing AUC. The TE asymmetry AUC of 0.50 is therefore a lower bound; its distance from the target AUC of 0.75 cannot be attributed to the TE asymmetry signal alone. Evaluating H4 conclusively requires exact per-electrode annotations from the HUP clinical team or validation on a dataset with published SOZ labels [23].

Recording duration and the MI-variance advance-time analysis. The 120-second ictal clips prevent evaluation of the MI temporal advance over signal variance, because the clips begin within the preictal period. Continuous long-term recordings (e.g., the MNI Open iEEG corpus) would enable this analysis as originally designed.

Lobe enrichment statistics. The frontal OR = 7.54 (p = 0.10) and MTL OR = 0.26 (p = 0.13) are directionally consistent with the predicted mechanisms but do not reach conventional significance. The classified sample (n = 46) is underpowered for Fisher's exact test; formal significance requires a larger cohort or reclassification using clinical SOZ lobe annotations rather than the participants.tsv surgical target field.

Electrode modality distribution. The cohort is skewed toward ECoG recordings (85%), with SEEG comprising only 15%. Whether the Type A/B prevalence generalises to SEEG-dominant cohorts (common in European centres) is an open question, because depth contacts sample different cortical volumes and are more susceptible to signals from non-target structures.

Single-site cohort. All analyses used a single-institution dataset. External replication on an independent iEEG cohort is required before the bifurcation finding can be claimed to generalise across recording centres, electrode configurations, and patient populations.

6. Conclusion

Two decades of conflicting evidence on whether preictal MI rises or falls before seizure onset now has a structural explanation. We showed that the group-level signed-rank test finds no evidence of uniform MI collapse (p = 0.92) because two biologically opposed dynamics coexist within the same cohort: frontal-onset patients exhibited preictal MI collapse (Cohen's $d \approx +0.99$, $n = 21$), consistent with information bottleneck compression in distributed frontal networks, while MTL-onset patients exhibited preictal MI amplification ($d \approx -2.63$, $n = 25$), consistent with hippocampal-entorhinal hypersynchrony. MI-weighted graph topology confirmed the bifurcation independently (16 fragmentation vs. 15 integration; $r = -0.60$), and MI-variance divergence uncovered an invisible preictal precursor in 6 of 16 frontal-onset patients that amplitude monitoring cannot detect. Mixed-cohort pooling mechanically produces a null result, explaining why prior studies have disagreed; the MI directionality phenotype, assignable from an interictal calibration window alone without labelled seizure data, opens a direct path toward personalised, zero-parameter closed-loop seizure detection.

Data and Code Availability

The HUP iEEG dataset is publicly available via NEMAR (OpenNeuro ID: ds004100) and Pennsieve Discover (dataset 179); no data access agreement is required. All analysis code (preprocessing, Kraskov MI estimator, rolling $\Phi(t)$ computation, graph metrics, variance comparator, TE asymmetry index) is published as Kaggle dataset `mdariffaysalnayem/ieeg-ib-core` and will be archived to a GitHub repository with a Zenodo DOI at time of submission. All analyses run on standard CPU hardware; no GPU is required.

Declarations

Competing interests. The authors declare no competing interests.
Funding. This work received no external funding.
Ethics. All data are de-identified and publicly available; no additional ethics approval was required.

A. MI Estimator Sensitivity Analysis

The Kraskov estimator has known finite-sample bias that increases as window length decreases. The 5-second window at 500 Hz provides 2,500 samples per channel pair (adequate for $k = 5$ nearest neighbours) and the bias is consistent across the recording, affecting Φ_{inter} and Φ_{pre} equally. Supplementary Figure S2 shows per-patient Cohen's d for window lengths from 1 to 30 seconds and $k \in \{3, 5, 7, 10\}$, demonstrating stability of the bifurcation classification across all tested parameters.

CRediT authorship contribution statement

Arif Faysal Nayem: Conceptualization, Methodology, Software, Formal analysis, Writing – Original draft.

References

[1] Kwan P, Brodie MJ. Early identification of refractory epilepsy. *New England Journal of Medicine*. 2000;342(5):314–319. doi:10.1056/NEJM200002033420503.
[2] Bergey GK, et al. Long-term treatment with responsive brain stimulation in adults with refractory partial seizures. *Neurology*. 2015;84(8):810–817. doi:10.1212/WNL.0000000000001230.
[3] Mormann F, Lehnertz K, David P, Elger CE. Mean phase coherence as a measure for phase synchronization and its application to the EEG of epilepsy patients. *Epilepsy Research*. 2003;53(1–2):173–185. doi:10.1016/S0920-1211(03)00002-0.
[4] Mormann F, Andrzejak RG, Elger CE, Lehnertz K. Seizure prediction: the long and winding road. *Brain*. 2007;130(2):314–333. doi:10.1093/brain/awl231.
[5] Schindler K, Leung H, Elger CE, Lehnertz K. Assessing seizure dynamics by analyzing the correlation structure of multichannel intracranial EEG. *Brain*. 2007;130(1):65–77. doi:10.1093/brain/awl316.
[6] Kramer MA, Cash SS. Epilepsy as a disorder of cortical network organization. *Neuroscientist*. 2012;18(4):360–372. doi:10.1177/1073858411411377.
[7] Bernabei JM, et al. Normative intracranial EEG maps of neural architecture and function. *Brain*. 2023;146(6):2248–2258. doi:10.1093/brain/awad007.
[8] Medina N, et al. Transient high-connectivity states as preictal biomarkers in focal epilepsy. *Journal of Neural Engineering*. 2025;22(4). doi:10.1088/1741-2552/adf097.
[9] Bullmore E, Sporns O. Complex brain networks: graph theoretical analysis of structural and functional systems. *Nature Reviews Neuroscience*. 2009;10(3):186–198. doi:10.1038/nrn2575.
[10] Sun J, et al. Causal spatio-temporal transfer entropy for epileptic seizure detection from EEG signals. *Entropy*. 2024;26(10):853. doi:10.3390/e26100853.
[11] Chen M, et al. Transfer entropy-based seizure onset zone localization in drug-resistant epilepsy. *Computer Methods and Programs in Biomedicine*. 2025;258:108483. doi:10.1016/j.cmpb.2024.108483.
[12] Van Mierlo P, et al. Functional brain connectivity from EEG in epilepsy: seizure-onset zone and seizure dynamics. *Progress in Neurobiology*. 2014;121:19–35. doi:10.1016/j.pneurobio.2014.06.004.

[13] Gramfort A, et al. MEG and EEG data analysis with MNE-Python. *Frontiers in Neuroscience*. 2013;7:267. doi:10.3389/fnins.2013.00267.

[14] Kraskov A, Stögbauer H, Grassberger P. Estimating mutual information. *Physical Review E*. 2004;69(6):066138. doi:10.1103/PhysRevE.69.066138.

[15] Watts DJ, Strogatz SH. Collective dynamics of 'small-world' networks. *Nature*. 1998;393(6684):440–442. doi:10.1038/30918.

[16] Rubinov M, Sporns O. Complex network measures of brain connectivity: uses and interpretations. *NeuroImage*. 2010;52(3):1059–1069. doi:10.1016/j.neuroimage.2009.10.003.

[17] Schreiber T. Measuring information transfer. *Physical Review Letters*. 2000;85(2):461–464. doi:10.1103/PhysRevLett.85.461.

[18] Tishby N, Pereira FC, Bialek W. The information bottleneck method. *Proceedings of the 37th Annual Allerton Conference on Communication, Control, and Computing*. 1999:368–377. arXiv:physics/0004057.

[19] Tishby N, Zaslavsky N. Deep learning and the information bottleneck principle. *IEEE Information Theory Workshop*. 2015:1–5. doi:10.1109/ITW.2015.7133169.

[20] Wibral M, Vicente R, Lizier JT. Directed information measures in neuroscience: a review and synthesis. In: Wibral M, Vicente R, Lizier JT, eds. *Directed Information Measures in Neuroscience*. Springer; 2014:3–36. doi:10.1007/978-3-642-54474-3_1.

[21] Trevelyan AJ, Schevon CA. How inhibition influences seizure propagation. *Neuropharmacology*. 2013;69:45–54. doi:10.1016/j.neuropharm.2012.06.017.

[22] Witter MP, Wouterlood FG (eds). *The Parahippocampal Region: Organization and Role in Cognitive Function*. Oxford University Press; 2002.

[23] Wagenaar JB, et al. A unified data-collection and analysis framework for invasive neural interfaces. *Journal of Clinical Neurophysiology*. 2015;32(3):235–239. doi:10.1097/WNP.0000000000000159.